On the next few pages are two (2) case studies.

Please **choose only one (1)** of the cases and completely answer **ALL** the questions that follow.

Reminder:
- Create one document for answers to both questions above
- Please name document for ease of identification.
    Use: YOUR FULL NAME_WEEK#_ ‘Assignment’
    Example: RALPHMARRA_WEEK1_Assignment

Week 1 - Case 1

Will the Coronavirus Pandemic Make Working from Home the New Normal?

As COVID-19 continued to spread around the globe, companies large and small started to make changes to the way they work, shuttering their offices and requiring most or all of their employees to work remotely from their homes.

- During the pandemic, ClearRisk, which offers integrated, cloud-based software solutions for claims, fleet, incident, and insurance certificate management had its entire staff working from home.
- Many large law firms, including Reed Smith, Baker McKenzie, and Nixon Peabody, closed offices and required work at home during the pandemic. The law firms emphasized that they could continue to serve clients despite office closings and remote work.
- OpenText Corp., a Canadian provider of enterprise information management products, plans to eliminate more than half of its 120 offices globally, with 2000 of its 15,000-person workforce working from home permanently.
- In mid-May 2020, Twitter Inc. notified employees that most of them could work from home indefinitely.

According to a recent MIT report, 34 percent of Americans who previously commuted to work stated that they were working from home by the first week of April 2020 due to the coronavirus outbreak. Prior to the pandemic, the number of people regularly working from home remained in the single digits, with only about 4 percent of the US workforce working from home at least half the time. However, the trend of working from home had been slowly gaining momentum thanks to advances in information technology for remote work and changes in corporate work culture. The coronavirus pandemic may mark a tipping point.

It's likely that many people who started working from home for the first time during the pandemic will continue to do so thereafter. New health guidelines about distancing will require some workplaces to expand to accommodate all their employees or to have a significant percentage of employees work permanently from home.

Information technologies driving these changes include broadband high-speed Internet connections, laptop computers, tablets, smartphones, email, messaging, and videoconferencing tools. As companies shift their work from face-to-face to remote, video conferencing is becoming the new normal for meetings. People are trying to have good conversations, share critical information, generate new ideas, reach consensus, and make decisions quickly on this platform.

Although less than ideal for face-to-face interactions, videoconferencing is becoming more powerful and affordable. There are many options, including Skype, Skype for Business, Zoom, Microsoft Teams, Amazon Chime, BlueJeans, Cisco's WebEx, GoToMeetings, and Google Meet. Some business people are using the same tools they do in their personal communications, such as FaceTime and Facebook Messenger. (FaceTime now supports group video chat with up to 32 people.)

Video conference software such as WebEx and BlueJeans appears designed for more corporate uses. Other software such as Microsoft's Skype and Zoom feels more consumer-friendly and easier to set up, with free or low-cost versions suitable for smaller businesses. Skype works for video chats, calls, and instant messaging and can handle up to 50 people in a single video call. Skype allows calls to be recorded in case someone misses a meeting. Skype also provides file-sharing capabilities, caller ID, voicemail, a split view mode to keep conversations separate, and screen share on mobile devices.

Up to 1,000 users can participate in a single Zoom video call, and 49 videos can appear on the screen at once. Zoom includes collaboration tools like simultaneous screen-sharing and co-annotation, and the ability to record meetings and generate transcripts. Users can adjust meeting times, select multiple hosts, and communicate via chat if microphones and cameras are turned off.

There are definite benefits to remote work: lower overhead, more flexible schedules, reductions in employee commuting time and attrition rates, and increases in productivity. (Many companies reported that productivity did not suffer when employees worked at home during the pandemic.) According to Global Workplace Analytics, a typical company saves about $11,000 per half-time telecommuter per year. Working remotely also poses challenges.

Not all employees have access to the Internet at home, and many work in industries that require on-site work. About 80 percent of American adults have high-speed broadband Internet service at home. However, according to a Pew Research Center study, racial minorities, older adults, rural residents, and people with lower levels of education and income are less likely to have in-home broadband service. In addition, one in five American adults access the Internet only through their smartphones. Employees with little children or small apartments find working at home more difficult.

Full-time employees are four times more likely to have remote work options than part-time employees. According to Global Workplace Analytics, a typical remote worker is college-educated, at least 45 years old, and earns an annual salary of $58,000 while working for a company with more than 100 employees.

Although email and text messaging are very useful, they are not effective tools for communication compared to the information exchange and personal connection of face-to-face conversations. Remote work also inhibits the creativity and innovative thinking that take place when people interact with each other face-to-face, and videoconferencing is only a partial solution. Studies have found that people working together in the same room tend to solve problems more quickly than remote collaborators, and that team cohesion suffers when members work remotely.

Sources: Lindsey Jacobson, "As Coronavirus Forces Millions to Work Remotely, the US Economy May Have Reached a 'Tipping Point' in Favor of Working from Home," CNBC, March 23, 2020; Rita Zeidner, "Coronavirus Makes Work from Home the New Normal," All Things Work, March 21, 2020; Dana Mattioli and Konrad Putzier, "The End of the Office," Wall Street Journal, May 16-17, 2020; Cate Pye, "Coronavirus: What Does the 'New Normal' Mean for How We Work?" *Computer Weekly*, April 3, 2020; Josh Lowy, "Overcoming Remote Work Challenges," *MIT Sloan Management Review*, April 9, 2020; Derek Thompson, "The Coronavirus Is Creating a Huge, Stressful Experiment in Working from Home," *The Atlantic*, March 13, 2020; Kevin Roose, "Sorry, but Working from Home Is Overrated," *New York Times*, March 10, 2020; Rani Molla, "This Is the End of the Office as We Know It," *Vox*, April 14, 2020.

Case Study Questions

1. Define the problem described in this case. What are the management, organization, and technology issues raised by this problem?
2. Identify the information technologies used to provide a solution to this problem. Was this a successful solution? Why or why not?
3. Will working from home become the dominant way of working in the future? Why or why not?

Week 1 - Case 2

UPS Competes Globally with Information Technology

United Parcel Service (UPS) started out in 1907 in a closet-sized basement office. Jim Casey and Claude Ryan—two teenagers from Seattle with two bicycles and one phone—promised the "best service and lowest rates." UPS has used this formula successfully for more than a century to become the world's largest ground and air package-delivery company. It's a global enterprise with more than 495,000 employees, 125,000 delivery vehicles, and 572 aircraft.

Today UPS delivers 5.5 billion packages annually in more than 220 countries and territories. The firm has been able to maintain leadership in smallpackage delivery services despite stiff competition from FedEx and the US Postal Service by investing heavily in advanced information technology. UPS spends more than $1 billion each year to maintain a high level of customer service while keeping costs low and streamlining its overall operations.

It all starts with the scannable bar-coded label attached to a package, which contains detailed information about the sender, the destination, and when the package should arrive. Customers can download and print their own labels using special software provided by UPS or by accessing the UPS website. Before the package is even picked up, information from the "smart" label is transmitted to one of UPS's computer centers in Mahwah, New Jersey, or Alpharetta, Georgia, and sent to the distribution center nearest its final destination.

Dispatchers at this center download the label data and use special routing software called ORION to create the most efficient delivery route for each driver that considers traffic, weather conditions, and the location of each stop. Each UPS driver makes an

average of 120 stops per day. In a network with 55,000 routes in the United States alone, shaving even one mile off each driver's daily route translates into big savings in time, fuel consumption, miles driven, and carbon emissions—as much as $50 million per year.

These savings are critical as UPS tries to boost earnings growth as more of its business shifts to less-profitable e-commerce deliveries. UPS drivers who used to drop off several heavy packages a day at one retailer now often make multiple stops scattered across residential neighborhoods, delivering one package per household. The shift requires more fuel and more time, increasing the cost to deliver each package.

The first thing a UPS driver picks up each day is a handheld computer called a Delivery Information Acquisition Device (DIAD), which can access a wireless cell phone network. As soon as the driver logs on, his or her day's route is downloaded onto the handheld. The DIAD also automatically captures customers' signatures along with pickup and delivery information. Package tracking information is then transmitted to UPS's computer network for storage and processing. From there, the information can be accessed worldwide to provide proof of delivery to customers or to respond to customer queries. It usually takes less than 60 seconds from the time a driver presses "complete" on the DIAD for the new information to be available on the web.

Through its automated package tracking system, UPS can monitor and even reroute packages throughout the delivery process. At various points along the route from sender to receiver, bar code devices scan shipping information on the package label and feed data about the progress of the package into the central computer. Customer service representatives are able to check the status of any package from desktop computers linked to the central computers and respond immediately to inquiries from customers. UPS customers can also access this information from the company's website using their own computers or mobile phones. UPS now has mobile apps and a mobile website for iPhone and Android smartphone users.

Anyone with a package to ship can access the UPS website to track packages, check delivery routes, calculate shipping rates, determine time in transit, print labels, and schedule a pickup. The data collected at the UPS website are transmitted to the UPS central computer and then back to the customer after processing. UPS also provides tools that enable customers, such as Cisco Systems, to embed UPS functions, such as tracking and cost calculations, into their own websites so that they can track shipments without visiting the UPS site.

UPS is now leveraging its decades of expertise managing its own global delivery network to manage logistics and supply chain activities for other companies. It created a UPS Supply Chain Solutions division that provides a complete bundle of standardized services to subscribing companies at a fraction of what it would cost to build their own systems and infrastructure. These services include supply chain design and management, freight forwarding, customs brokerage, mail services, multimodal transportation, and financial services in addition to logistics services.

UPS technology and business services are helpful to businesses of all sizes, including small start-ups. Fondarific is a Savannah-based company that manufactures and sells fondant icings for decorating wedding cakes and childrens'cakes. UPS made it possible for Fondarific to grow rapidly when international sales took off. UPS set up a class in exporting to teach Fondarific how to manage international sales and logistics and how to use its WorldShip global shipping software for UPS package and freight services. UPS also showed the company how to integrate shipping systems with Quickbooks accounting software and inventory software.

UPS provides both financial and shipping advice and services to 4Moms, a Pittsburgh-headquartered company with 80 employees that makes innovative baby products using consumer technology. 4Moms uses UPS Trade Direct, which enables companies to reduce freight and inventory costs by bypassing distribution centers and shipping their goods directly to retailers. The UPS Cargo Finance service helps 4Moms manage the cost of inventory as it is shipped around the world.

Sources: www.ups.com, accessed April 27, 2020; Sean Galea-Pace, "UPS Supply Chain Introduces Smart Warehouse Technology," *Supply Chain Digital*, April 24, 2020; and Bloomberg, "UPS Sees Payoff From $20Bn Tech Bet," *SupplyChainBrain*, April 24, 2019.

Case Study Questions

1. What are the inputs, processing, and outputs of UPS's package tracking system?
2. What technologies are used by UPS? How are these technologies related to UPS's business strategy?
3. What strategic business objectives do UPS's information systems address?
4. What would happen if UPS's information systems were not available?